IQP Generative Models for Strong Gravitational Lensing Images

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Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed hard to sample classically [5, 6], while the expectation values a maximum-mean-discrepancy (MMD) loss needs stay computable, so such a Born machine can be trained classically and deployed on quantum hardware [24]. A recent correlation–complexity map is proposed to indicate which data suit them, and DeepLense strong-lensing images sit at the most IQP-compatible position among the encodings we measured. The reference protocol's frozen core with sequential adaptation yields only noise, which joint training repairs, and a learned linear map then loses coherence past $M{=}6$, so we let a small non-linear map carry the latent. At eleven training images the resulting 10-qubit machine matches a DCGAN of six times as many parameters on both generation metrics within the seed spread, is ahead of it on a flux-aligned variant and of an RBM on the two MMD^2 columns while level with it on the histogram metric, and projects unseen images at correlation 0.901; at 100 images the DCGAN is ahead and that correlation reaches 0.977. Every circuit is simulated exactly at $n = 10$ qubits over five seeds, so we claim no quantum advantage, only an early test of whether such a model could be worth deploying where training data are scarce.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard [30] transform of a phase function. Exact or multiplicatively approximate classical sampling from them would collapse the polynomial hierarchy [5], and that hardness is conjectured to hold on average under additive error [6], a bounded motivation, since noise makes such circuits simulable in polynomial time [23], classical algorithms keep improving [18] and trainability itself pulls towards simulability [7]. The same structure keeps an MMD loss's expectation values computable [25], as it does for other encodings built to stay simulable [17], so such a model can be *trained classically and deployed on quantum hardware* [24], in the line of Born machines trained by an MMD loss [16, 3, 9].

Whether real data sit where that is useful is empirical, and a recent proposal [15] approaches it with two ingredients. (i) A *correlation–complexity map* placing a dataset at I_{QCLI}, the Jensen–Shannon divergence between its Walsh order spectrum, the Fourier power at each Hamming weight, and the i.i.d. binomial baseline, and $I_{\mathrm{CCI}} \in [0, 1]$, the share of its correlation the best pairwise tree misses [31, 26, 8], the high-I_{QCLI}, high-I_{CCI} corner being where IQP circuits should do well. (ii) A latent-adaptation protocol (Algorithm 1 of that paper's v2, a phase core trained on one anchor plus the first 50 phases in Hamming order, randomly initialised and adapted sample by sample), demonstrated at $n = 18$ qubits on 2D turbulence [13], one of several datasets the map covers. A follow-up initialises that same fixed block from a physics-informed surrogate [14], where we instead learn the directions.

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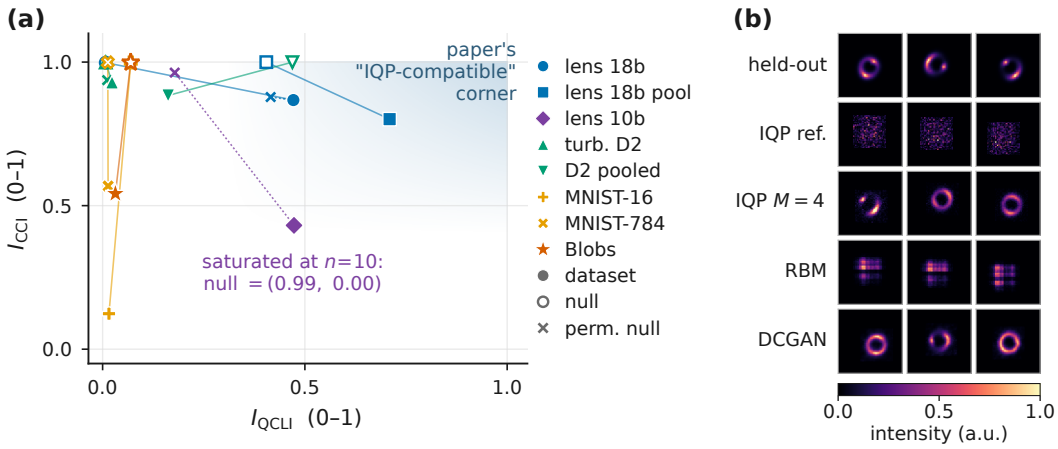


Figure 1: **(a)** Correlation–complexity map, datasets filled, i.i.d.-uniform nulls open, permutation nulls as crosses (ours); lens 18b and 18b pool are the per-image and pooled 18-bit encodings, lens 10b the crop pmf, turb. D2 the reference turbulence data. **(b)** First three unconditional samples per model, $N{=}100$, no selection.

We take both to simulated dark-matter strong-lensing images from DeepLense [1, 2] and contribute three things. We recompute the map on lens crops, which land at the most IQP-compatible position we measured, and show with a permutation null that there the index is largely determined by the intensity histogram, so a favourable position is not by itself evidence of IQP-suitable structure. We trace why the reference latent protocol produces noise on static images and replace its latent map by a non-linear one. We then compare against an RBM and a DCGAN under two intensity conventions, and report a replication on the turbulence data in App. C.

2 Setup

Model. Every arm is a 10-qubit IQP Born machine over $1{,}024$ Walsh phases θ, one per qubit subset and $1{,}023$ of them effective since the empty subset is a global phase, differing only in how a per-image latent enters. The reference protocol freezes the first 50 of those phases in Hamming order and adapts them sample by sample, which on static crops leaves the other training images poorly reconstructed (Sec. 3.2), so we make that map trainable, first as a linear map of M directions,

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M\times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with each row of Φ normalised in L^2. That map degrades by $M{=}6$ on these crops, we suspect once the $\|z\|$ the images demand leaves the radius over which the Walsh phases interfere coherently, though we do not isolate the cause; our main arm replaces $\Phi^\top z$ by a one-hidden-layer $\tanh$ map g of width $H{=}32$, the z_i trained jointly with g under a small L^2 penalty. The main objective is the flat MMD (uniform Walsh weights, estimable from hardware shots), with the three-bandwidth Gaussian kernel of [24], the *paper kernel* of the tables, kept as a secondary arm, and θ^0 is warm-started by $1{,}000$ exact-KL steps, which a flat-MMD warm start replaces at no cost (App. C).

Decoding, baselines and metrics. The trained pmf fixes only the relative intensities, so it is pasted into a 64×64 frame at a total intensity S and at an offset drawn from the training crops' own offsets. The reference convention gives every frame the training-set mean S, leaving the generated flux with zero variance where real crops spread by about a quarter of their mean, so we draw S per image from a kernel density estimate of the N training totals, with no retraining; App. A repeats everything under the fixed S. The baselines are a restricted Boltzmann machine (RBM) [28, 12, 29] and a deep convolutional generative adversarial network (DCGAN) [22], both early-stopped on 200 validation images while the IQP arms run a fixed $4{,}000$ steps, which can only favour them. We score with the Jensen–Shannon divergence between intensity histograms (PDF–JS) and Conv2D-MMD^2 in a fixed random encoder's feature space [11, 4], add a post-hoc flux-aligned Conv2D-MMD^2 rescaling every frame to the held-out mean total since the arms do not agree on how flux is set, and report a held-out

projection correlation, the best of eight latent restarts by the fit's own loss. Bins, scale and bandwidth are set on the held-out data alone, so no metric moves with the arm scored, and every arm generates 200 images against the same 200 held-out frames at $N \in \{11, 100\}$, every number a mean $\pm$ sample sd over five seeds beside a real-vs-real floor between two disjoint real sets. The N training images are fixed across seeds, so that spread is initialisation and sampling variance alone. A lead in sd means the difference of two means in units of the baseline's seed sd, descriptive and not a significance test.

Departures from the reference setup. Four choices differ, each following from the data. The reference snapshots interpolate to unseen times, whereas the DeepLense `Model II` images we use [19] are static, so we generate unconditionally and project onto held-out images. Its 18-bit $(x, y, \mathrm{intensity})$ encoding suits a field of tuples, whereas a lens crop's raw intensity axis carries barely one bit, so we centroid-align each image and read its central 32×32 crop, 99.4% of the light, as one probability mass function (pmf) over $1{,}024$ cells at $n = 10$ (App. A). It fixes the latent block, whereas we learn the map and train the N images jointly, and it estimates its MMD from samples where a 10-qubit pmf is exact.

3 Results

3.1 Where lens crops sit on the correlation–complexity map

Table 1 recomputes the map (Fig. 1(a)). Encoded as the reference paper encodes turbulence, lens images sit far above it in I_{QCLI} and inside the corner the map calls IQP-compatible, yet a permutation null keeping each image's intensity histogram comes within 14% of the lens value and a factor of two of the much smaller turbulence value, so there the index largely tracks the histogram. Both indices depend on K, the codes per image, so a row reads only against its own-K null; at the protocol's own $N_t = 5 \times 10^4$ samples per snapshot [15] the turbulence I_{QCLI} falls to 0.091 against a null of 0.098. The 10-bit crop pmf, the encoding we model, is the one that separates from its null in both coordinates, but sits at $I_{\mathrm{CCI}} = 0.43$, outside the high-I_{CCI} corner, so its favourable I_{QCLI} motivates the encoding, not the choice of an IQP over a pairwise model.

3.2 From the reference protocol to a non-linear latent map

Run sequentially on our own 50-phase block (App. C), Algorithm 1 produces noise (Fig. 1(b), second row). Neither representability nor the block is the cause, since the full circuit fits a *single* aligned crop to correlation 0.99997 ± 0.00001 and training the same 50 phases jointly with the core across all N images (the *fixed basis, joint* row of Table 2) already recovers most of the gap. What fails is the frozen core with sequential per-image adaptation, since with the core frozen on one anchor the 50 fixed coordinates reconstruct every other training image at only 0.44 ± 0.04; the block size of 50 is not swept. Equation (1) repairs the rest but introduces a width limit of its own. On a 20-image sweep the learned linear map holds a held-out projection correlation of 0.906 ± 0.002 at $M{=}2$, falls to 0.738 ± 0.067 at $M{=}6$ and collapses to 0.218 ± 0.121 at $M{=}8$. The non-linear map removes that collapse, holding 0.9368 to 0.9519 across the sweep, a little above that split's nearest-training-image reference of 0.9339; the two sweeps come from separate drivers differing in optimiser schedule and initialisation, both scored on the same 20 held-out frames, so a small offset between them is not a model difference. It does not buy a large useful width, being flat to $M{=}4$ and degrading from $M{=}8$. We report it at $M{=}2$, where the held-out projection correlation is 0.901 at $N{=}11$ and 0.977 at $N{=}100$, against 0.892 and 0.970 for projecting the nearest training image and 0.838 and 0.887 for a single zero-initialised restart.

Table 1: Correlation–complexity map, each permutation null keeping the value histogram of the row above, five seeds; ± 0.0000 marks a seed spread below 5×10^{-5}.

Dataset and encoding	K	I_{QCLI}	I_{CCI}
Lens, 18-bit $(x, y, \mathrm{intensity})$ tuples	4,096	0.4715 ± 0.0000	0.8677 ± 0.0000
permutation null	4,096	0.4154 ± 0.0000	0.8783 ± 0.0001
Turbulence D2, 18-bit tuples	4,096	0.0232 ± 0.0002	0.9285 ± 0.0008
permutation null	4,096	0.0114 ± 0.0000	0.9371 ± 0.0009
Lens, 10-bit crop pmf	exact pmf	0.4731 ± 0.0000	0.4313 ± 0.0000
permutation null	exact pmf	0.1782 ± 0.0000	0.9634 ± 0.0004

Table 2: Lens crops under the sampled intensity, five seeds, PDF–JS in 10^{-3} and both MMD^2 columns in 10^{-2}. Lower is better; **bold** marks a best mean whose gap to *every* other arm exceeds the sum of the two seed sds, so a column without bold carries no win the seed spread supports. Params are totals at N=11 / N=100 and App. C gives † and the parameter convention.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD$^2_{\text{al}}$	PDF–JS	MMD^2	MMD$^2_{\text{al}}$
IQP, reference protocol (Alg. 1)†	1,524 / 5,974	6.5 ± 0.8	14.8 ± 1.6	22.5 ± 2.2	7.7 ± 0.4	14.7 ± 0.3	26.6 ± 0.7
IQP, fixed basis, joint	1,524 / 5,974	1.8 ± 0.4	8.7 ± 1.4	12.5 ± 1.3	1.1 ± 0.2	5.4 ± 0.4	8.5 ± 1.5
IQP, linear map M=4 (flat MMD)	5,164 / 5,520	2.8 ± 0.1	5.6 ± 1.9	2.7 ± 0.2	2.6 ± 0.2	2.8 ± 0.5	2.6 ± 0.4
IQP, linear map M=4 (paper kernel)	5,164 / 5,520	1.0 ± 0.05	6.3 ± 1.5	7.4 ± 0.6	1.5 ± 0.2	5.9 ± 0.4	10.0 ± 0.8
IQP, non-linear map M=2 (flat MMD)	33,910 / 34,088	1.2 ± 0.07	5.2 ± 1.8	**2.2** ± 0.2	1.7 ± 0.2	2.4 ± 0.5	1.8 ± 0.2
IQP, non-linear map M=2 (paper kernel)	33,910 / 34,088	0.75 ± 0.07	5.4 ± 1.5	5.1 ± 0.3	0.99 ± 0.04	4.2 ± 0.4	6.6 ± 0.6
RBM [29]	11,510 / 15,960	1.9 ± 1.0	7.9 ± 1.9	11.1 ± 3.9	1.1 ± 0.2	5.1 ± 0.8	8.1 ± 2.2
DCGAN [22]	199,837	1.1 ± 0.6	5.1 ± 1.5	3.9 ± 1.0	**0.55** ± 0.12	2.2 ± 0.3	2.1 ± 0.4
Real-vs-real floor	n/a	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2

3.3 Comparison with classical baselines

In Table 2 the eleven-image regime is where the IQP arms are competitive. There the non-linear arm matches the DCGAN on PDF–JS and raw Conv2D-MMD^2 within seed noise (-0.13 and -0.04 of the DCGAN's own sd) and is ahead on the flux-aligned column ($+1.57$ sd), and against the RBM it is ahead on the two MMD^2 columns by 1.40 and 2.27 of the RBM's sd with PDF–JS inside one sd (0.75), at a sixth of the DCGAN's parameters and three times the RBM's. Under the paper kernel the same map holds the best mean PDF–JS, 0.75×10^{-3} against the DCGAN's 1.1 ± 0.6, which leaves the two inside one sd, and trades that for a worse flux-aligned column. No arm is ahead of the DCGAN on all three columns by more than a seed sd at either size, and at 100 images it is ahead on PDF–JS and raw Conv2D-MMD^2, the flux-aligned 1.8 ± 0.2 sitting inside one sd of its 2.1 ± 0.4.

Three properties bound how the table reads. The flux-aligned column is invariant to the intensity convention, the largest paired difference being 6×10^{-9} over the 325 runs carrying both. That is invariance, not independence from the raw column, so the two MMD^2 columns measure shape, and shape plus flux, not two separate things. The sampling rule is also not neutral between arms, improving every IQP arm's raw Conv2D-MMD^2 by 22 to 39% at N=11 while making the RBM's 20% worse, and under the fixed intensity the RBM leads the non-linear arm there (6.5 ± 0.8 against 8.5 ± 0.2, App. A), so the raw-column standing against the RBM is a property of the sampled decode and not of the model. Sampling fixes the shape of the generated flux distribution and not its location, the eleven training frames being 9% dimmer than the held-out set, which is why the raw column stops there.

Data scaling over $N \in \{10, \ldots, 100\}$ and per-class retraining are reported in App. B.

4 Discussion and limitations

At eleven training images a 10-qubit IQP Born machine with a learned non-linear latent map is competitive with an RBM and a DCGAN, matching the DCGAN on both generation metrics within the seed spread at a sixth of its parameters and holding the lowest flux-aligned mean; that standing does not survive to 100 images. Nothing here is evidence of quantum advantage, since every circuit is simulated exactly at $n = 10$ qubits, and any deployed margin is bounded by connectivity and noise [21]. Generated frames sit farther in L_2 from their nearest training image than real held-out frames do (4.21 ± 0.10 against 2.90 at N=11), so the arm is not memorising. Two steps are not shot-estimable as run, the exact-KL warm start, replaced in App. C within one seed sd, and the projection, which optimises z against the exact pmf; and at $n = 10$ we parameterise all 2^n Walsh phases where the scaling up would need them sparsified [15], untested here. Only the model fit sees N images, while bandwidths, floors and the baselines' early stopping use hundreds to thousands of real frames, so this is a small-training-set study, not a small-data pipeline; alignment and the crop leave translation and the outer 0.6% of the light outside the task, floor included. Five seeds are few, the DCGAN adds up to 57% of spread at one seed and alone is not bit-reproducible, the sampled intensity moves the raw column unequally, everything rests on one simulated dataset, the metrics do not test substructure, and no destroyed image is scored as an upper reference.

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A Pipeline and the fixed-S convention

Rerunning Table 2 with the single training-mean intensity every pmf-based arm published under rebuilds the published frames exactly from the same runs, so the two decodes are paired. Sampling improves every IQP arm's raw Conv2D-MMD^2 at $N{=}11$ by 0.033 to 0.041 (paired sd 0.012 to 0.020) while making the RBM worse by 0.013 ± 0.017, which is why the RBM leads the non-linear arm there under the fixed intensity (6.5 ± 0.8 against 8.5 ± 0.2); at $N{=}100$ it improves every arm, the PDF–JS effect is at most 7×10^{-4} and mixed in sign, and the flux-aligned effect is below 10^{-8}. The full table is in the released seed tables.

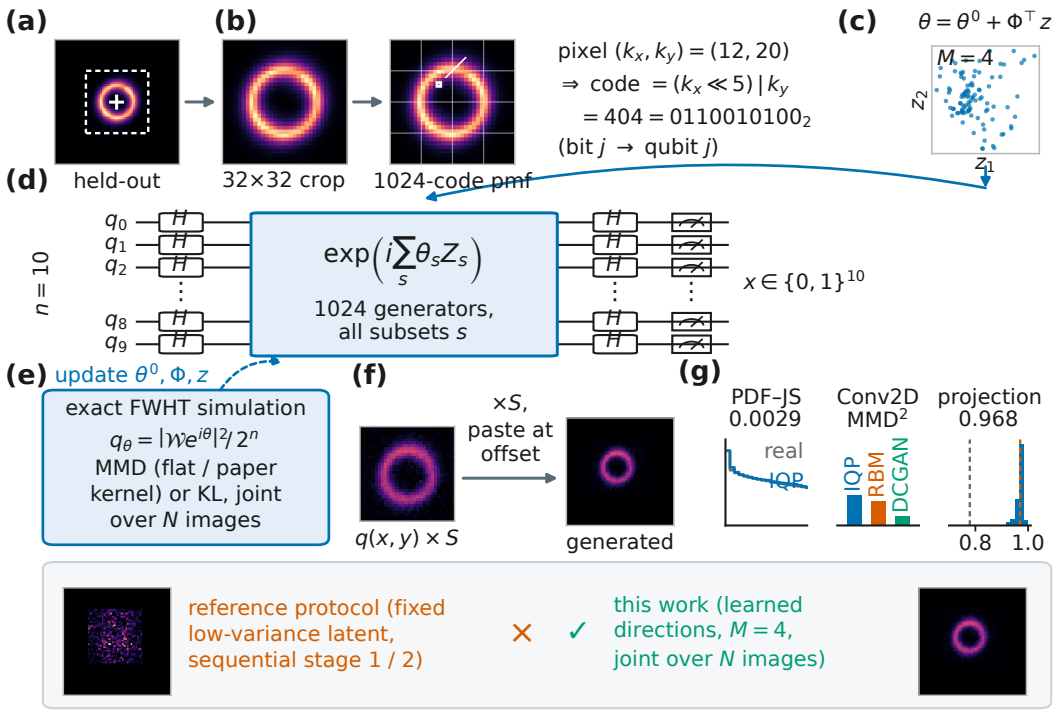


Figure 2: Pipeline. **(a)** A held-out image and the crop window. **(b)** The aligned crop as a pmf over 1,024 cells, the pixel $(12, 20)$ carrying code 404, bit j addressing qubit j. **(c)** Learned latents at $M = 4$. **(d)** The circuit. **(e)** Exact simulation, drawn for Eq. (1); the main arm holds $\theta = \theta^0 + g(z)$. **(f)** Decoding at S. **(g)** Metrics for the linear $M{=}4$ arm.

Table 3: Per class, each arm retrained on that class alone and scored against that class's own 200 held-out frames; sampled intensity, five seeds, units of Table 2. A column reads down a class, not across classes. The real-vs-real floors are $0.03/0.36/0.31$ for axion, $0.03/0.30/0.32$ for cdm and $0.04/0.45/0.45$ for no sub, in the same units and order.

		non-linear $M{=}2$			linear $M{=}4$		
Class	N	PDF–JS	MMD^2	MMD$^2_{\text{a1}}$	PDF–JS	MMD^2	MMD$^2_{\text{a1}}$
axion	11	1.1 ± 0.10	4.6 ± 0.6	2.9 ± 0.4	3.4 ± 0.20	4.7 ± 0.7	2.8 ± 0.5
axion	100	1.8 ± 0.17	2.9 ± 0.2	1.7 ± 0.2	2.6 ± 0.11	2.7 ± 0.2	2.1 ± 0.2
cdm	11	0.94 ± 0.05	4.4 ± 0.2	2.7 ± 0.3	2.4 ± 0.09	4.3 ± 0.3	2.6 ± 0.5
cdm	100	1.7 ± 0.06	2.1 ± 0.3	1.5 ± 0.3	2.7 ± 0.25	2.3 ± 0.4	1.8 ± 0.3
no sub	11	0.98 ± 0.10	3.1 ± 0.9	2.4 ± 0.6	2.6 ± 0.22	3.5 ± 0.8	2.6 ± 0.1
no sub	100	1.7 ± 0.10	2.3 ± 0.4	2.1 ± 0.3	2.7 ± 0.31	2.7 ± 0.4	2.9 ± 0.3

B Latent sweeps and per-class results

Hidden width matters less than latent width, everything from $H{=}16$ up being inside a seed sd, and the latent-width sweep is flat over $M \leq 4$ before degrading at $M{=}8$ and collapsing at $M{=}16$; both full sweeps are in the released seed tables. Across $N \in \{10, 30, 50, 70, 90, 100\}$ the non-linear arm's raw Conv2D-MMD^2 falls from 0.062 ± 0.016 to 0.024 ± 0.005 and the RBM's from 0.091 ± 0.017 to 0.051 ± 0.008, the DCGAN starting ahead at $N{=}10$ (0.046 ± 0.017) and ending level, while on the flux-aligned column it runs 0.026, 0.023, 0.020, 0.016, 0.019, 0.018 against the RBM's 0.077 to 0.127, lowest at every N but only at $N{=}10$ and $N{=}70$ by more than a seed sd. Retrained inside one class (Table 3, Fig. 3) each class can be generated on its own, which is not the same as the model separating them. The mean $\|z\|$ at $N{=}11$ is 0.101 at $M{=}2$ and 0.120 at $M{=}8$ while the affine residual of g falls from 0.70 to 0.17, so g is most nearly linear where it fails.

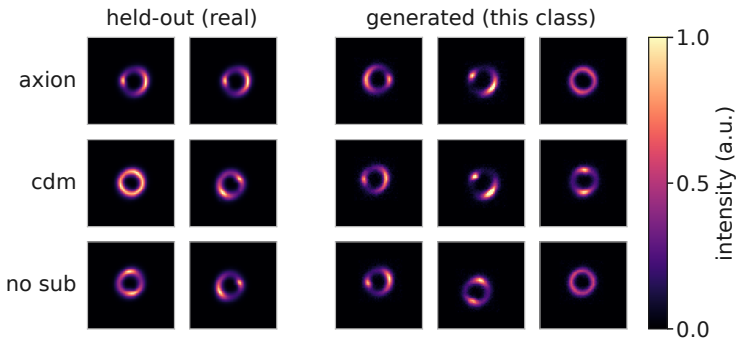


Figure 3: One row per substructure class at $N{=}100$, sharing one 0 to 1 intensity scale. Left, two held-out frames of that class; right, three unconditional samples from the non-linear $M{=}2$ arm retrained on that class alone, first two and first three in index order, no selection, seed 0.

C Turbulence replication, training and evaluation details

On that paper's own dataset (2D cylinder-wake turbulence D2 [13], $n = 18$ qubits, 11 anchor snapshots, five seeds) the result is **not reproduced** under our pre-registered convention, the RBM (PDF–JS 0.217 ± 0.007) and a 1.2M-parameter DCGAN (0.045 ± 0.062) both scoring better than our 31k-parameter IQP (0.272 ± 0.004) on *both* metrics. Under a last-write-wins aggregation, never specified in that paper [15], our PDF–JS moves towards the published values without reaching them (0.0324, 0.0473 against 0.0238, 0.0221), and the RBM still leads. Conv2D-MMD^2 there does not rank a destroyed image below a trained one, pixel-shuffled ground truth scoring 0.7045 ± 0.0095, better than every model in that paper's Table 2; that paper notes itself that its baselines are not exhaustive and that this metric is no substitute for spectral diagnostics [15].

Arms, baselines and protocol. The † arm runs Algorithm 1 of v2 [15] on our own latent block, the 50 subsets of Hamming weight at most 2 whose Walsh coefficients vary most over a disjoint pool of 2,000 images, where the reference block is the first 50 in Hamming order; *fixed basis, joint* is that block trained jointly with the core across all N images. Params are totals at that N, shared (θ^0 included) plus N per-image latents, so the fixed-basis arms hold $974 + 50N$, the learned-direction arms $5{,}120 + MN$ and the RBM $10{,}960 + 50N$; the DCGAN entry is its generator alone, its discriminator adding 53,497. The RBM has 10 visible and 1,000 hidden units on the 10-bit code, each image owning its visible biases and the first 40 hidden biases, and ran 160k or 80k steps; the DCGAN builds the crop from a 128-dimensional prior through four transposed convolutions, ran 5k to 30k steps and learns its own intensity. The paper kernel is the three-bandwidth Gaussian MMD of [24] at $\sigma = \{0.989, 1.729, 2.236\}$, with σ_1 set so the average Pauli weight is 2, σ_3 the square root of the median heuristic and σ_2 their root mean square, since the protocol we replicate publishes none for this encoding. The Conv2D-MMD^2 bandwidth is 0.005547 raw and 0.002558 flux-aligned, fixed once on the held-out features and shared by every table except the per-class rows; the 20 floor draws come from the 8,710 validation images the held-out split leaves; and the Silverman bandwidth [27] of the intensity estimate is 14.64 at $N{=}11$ and 10.36 at $N{=}100$. The baselines' validation rule applied to the IQP arms selects 2k to 16k steps and moves every Table 2 cell it changes toward better IQP values without altering a comparison, so our pre-registered 4,000 steps is the conservative choice, and at 20,000 fixed steps both baselines are much worse, so their early stopping is load-bearing and granted. Of three ablations none moves a generation metric by more than a seed sd. A flat-MMD warm start of θ^0, making every stage shot-estimable, leaves the flux-aligned column at 0.0231 against 0.0225 at $N{=}11$; picking the projection restart by correlation rather than by fit loss moves the correlation by 0.0003; and two restarts match eight (0.901, 0.977) where one zero-initialised restart reaches 0.838 and 0.887. The projection references at $M{=}2$ are 0.892 and 0.970 for the nearest training image and 0.765 and 0.779 for the mean crop. Our Conv2D-MMD^2 is the kernel inception distance [4] with an untrained encoder, we do not use the purpose-built generalisation metrics of [10], and classical generation of cosmological images is established [20]. As Wasserstein-1 distance in ring radius and the two ellipticity components the main arm is at $0.82/0.33/0.35$ at $N{=}11$ against $3.23/0.46/0.24$ (RBM), $1.67/0.62/0.40$ (DCGAN) and $0.10/0.17/0.10$ between two real sets; at $N{=}100$ the DCGAN is closest, $0.40/0.28/0.15$ against $1.01/0.31/0.14$. Retraining the same

DCGAN configuration at the same seed moves its raw Conv2D-MMD^2 by 5 to 57% and its PDF–JS by 2 to 74%, which is what we mean by the one arm that is not bit-reproducible.

D Qubit count, the reference paper's Fig. 4, and the departures in full

The qubit count has to match the effective resolution, a choice the reference paper does not face at $n = 18$ on a field of tuples. At $n = 8$ the 16×16 pmf averages the ring away and the fixed bilinear upsampling cannot put it back, so the flux-aligned error is more than twice the $n = 10$ value; at $n = 12$ about three quarters of the $4{,}096$ cells carry background rather than lens light, the arm learns that background and becomes unstable across seeds, its N=100 PDF–JS reaching 92 ± 71 with a spread as large as the mean. We therefore work at $n = 10$.

Table 4: Qubit ladder at N=11, five seeds, units of Table 2. The flux-aligned column is invariant to the intensity convention (App. A) and compares across rows; the other two are under the fixed intensity except the $n = 8$ row, marked *. The latent map differs by row, so this is a ladder of encodings, not a sweep of n alone.

n	Encoding	Latent map	Params	PDF–JS	MMD^2	MMD$^2_{\mathrm{al}}$
8	16×16 pmf, bilinear upsampling	MLP H=8, M=2	2,350	$1.7 \pm 0.05^*$	$8.0 \pm 1.7^*$	5.1 ± 0.3
10	32×32 aligned crop pmf	MLP H=32, M=2	33,910	1.0 ± 0.1	8.5 ± 0.2	$\mathbf{2.3} \pm 0.2$
12	whole 64×64 frame as one pmf	linear, M=4	20,524	43 ± 30	23.3 ± 8.8	18.4 ± 11.3

Re-running that paper's Fig. 4 over five SPSA seeds, its positive low-I_{QCLI} coupling ($I_{\mathrm{QCLI}} \leq 0.3$) reproduces at every size and in every seed, Spearman $+0.92 \pm 0.10$, $+0.91 \pm 0.04$ and $+0.96 \pm 0.01$ at $n = 8$, 12, 16, while above that range the sign of the global coefficient changes from seed to seed.

Table 5: Every place our protocol departs from the one we replicate, with the property of the data that drove it; none of them is a tuning result.

	Reference protocol	This paper	Property of the data
Encoding	18-bit $(x, y, \mathrm{intensity})$ tuples, many per snapshot	one 10-bit pmf over $1{,}024$ cells of an aligned 32×32 crop	the raw intensity axis carries barely one bit and 99.4% of the light is inside the crop
Latent	a fixed block of 50 phases, the first in Hamming order	a learned map, linear $\Phi^\top z$ or a $\tanh$ $g(z)$	on static images fixed directions cross-fade pixels instead of moving ring structure
Training	core fitted on one anchor, then each sample adapted in sequence	core and all N latents trained jointly, $4{,}000$ Adam steps	with at most 100 static frames there is no stream to adapt along
Objective	multi-bandwidth Gaussian MMD estimated from samples	flat MMD as the main arm, that kernel kept as a secondary one	at $n = 10$ the pmf is exact, so sampling noise would be a choice
Flux	every decoded frame carries the training-set mean total	S drawn per image from a density estimate of the N totals	real crops spread by about a quarter of their mean, which one S cannot reproduce
Evaluation	held-out times, that paper's own metrics	200 held-out frames fixed across arms and seeds, bins and bandwidth set on them alone	the images are static, so time cannot be held out
Baselines	compared at a fixed budget	RBM and DCGAN early-stopped on 200 validation frames, IQP arms not	the baselines converge at very different step counts, so one budget would favour us